

GEOGRAPHY

PHYSICAL GEOGRAPHY



GEOMORPHOLOGY

MODULE - 27

- **GEOMORPHIC PROCESSES**
WEATHERING- PHYSICAL, CHEMICAL
AND BIOLOGICAL (PART-1)

WEATHERING



- 1 Weathering is action of elements of weather and climate over Earth materials.
- 2 Weathering is defined as **mechanical disintegration and chemical decomposition of rocks** through the actions of various elements of weather and climate.
- 3 There are a number of processes within weathering **which act either individually or together to affect the Earth materials** in order to reduce them to fragmental state.

WEATHERING

3 As very little or no motion of materials takes place in weathering, it is an in-situ or on-site process.

4 Weathering processes are conditioned by many complex geological, climatic, topographic and vegetative factors where Climate is of particular importance.

5 There are three major groups of weathering processes

- i. Chemical
- ii. Physical or mechanical
- iii. Biological weathering processes

6 Very rarely does any one of these processes ever operate completely by itself, but quite often a dominance of one process can be seen.

WEATHERING

1

CHEMICAL WEATHERING PROCESSES

- 1 A group of weathering processes viz; solution, carbonation, hydration, oxidation and reduction act on the rocks to **decompose, dissolve or reduce them to a fine clastic state** through chemical reactions by oxygen, surface and/or soil water and other acids.
- 2 **Water and air (Oxygen and Carbon dioxide) along with the heat** must be present to speed up all chemical reactions.
- 3 Over and above carbon dioxide present in the air, **decomposition of plants and animals increase the quantity of carbon dioxide underground.**
- 4 These chemical reactions on various minerals are very much similar to the chemical reactions in a laboratory.

PHYSICAL WEATHERING VS CHEMICAL WEATHERING



WEATHERING

1

CHEMICAL WEATHERING PROCESSES



SOLUTION

1

When something is dissolved in water or acids, the water or acid with dissolved contents is called Solution.

2

On coming in contact with water many solids disintegrate and mix-up as suspension in water. Soluble rock forming minerals like nitrates, sulphates and potassium etc. are affected by this process.

3

So, these minerals are easily leached out without leaving residue in rainy climates and accumulate in dry regions.

WEATHERING

1

CHEMICAL WEATHERING PROCESSES



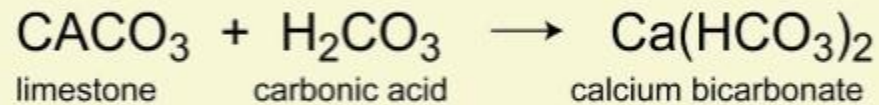
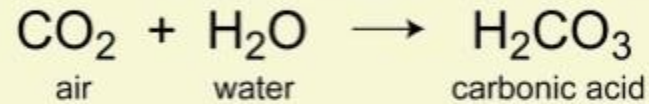
SOLUTION

- 4 Minerals like calcium carbonate present in limestones are soluble in water containing carbonic acid (formed with addition of carbon dioxide in water) and are carried away in water as solution.
- 5 Carbon dioxide produced by decaying organic matter along with soil water greatly aids in this reaction.
- 6 Common salt (Sodium Chloride) is also a rock forming mineral and is susceptible to this process of solution.

WEATHERING

1

CHEMICAL WEATHERING PROCESSES



CARBONATION

1

Carbonation is the reaction of carbonate and bicarbonate with minerals and is a common process helping the breaking down of feldspars and carbonate minerals.

2

Carbon dioxide from the atmosphere and soil air is absorbed by water, to form carbonic acid that acts as a weak acid.

3

Calcium carbonates and magnesium carbonates are dissolved in carbonic acid and are removed in a solution without leaving any residue resulting in cave formation.

WEATHERING

1

CHEMICAL WEATHERING PROCESSES



Gypsum

HYDRATION

- 1 Hydration is the **chemical addition of water**. Minerals take up water and expand; this expansion **causes an increase in the volume of the material itself or rock**.
- 2 Calcium sulphate takes in water and turns to gypsum, which is more unstable than calcium sulphate.
- 3 Continued repetition of the process of hydration **causes fatigue in the rocks and may lead to their disintegration**.

WEATHERING

1

CHEMICAL WEATHERING PROCESSES



HYDRATION

4 Many clay minerals swell and contract during wetting and drying and a repetition of this process results in cracking of overlying materials.

5 Salts in pore spaces undergo rapid and repeated hydration and help in rock fracturing.

6 The volume changes in minerals due to hydration will also help in physical weathering through exfoliation and granular disintegration.

WEATHERING

1

CHEMICAL WEATHERING PROCESSES



Natural chemical weathering of sulfide rocks feeds acid and iron into this spring.

OXIDATION AND REDUCTION

1

In weathering, oxidation means a combination of a **mineral with oxygen** to form **oxides or hydroxides**.

2

Oxidation occurs where there is ready **access to the atmosphere and oxygenated waters**. The minerals most commonly involved in this process are iron, manganese, sulphur etc.

3

In the process of oxidation rock breakdown occurs due to the disturbance caused by addition of oxygen.

WEATHERING

1

CHEMICAL WEATHERING PROCESSES

OXIDATION AND REDUCTION

- 4 Red colour of iron upon oxidation turns to brown or yellow.



- 5 When oxidised minerals are placed in an environment where oxygen is absent, reduction takes place.

- 6 Such conditions exist usually below the water table, in areas of stagnant water and water logged ground.

- 7 Red colour of iron upon reduction turns to greenish or bluish grey.

- 8 These weathering processes are inter-related. Hydration, carbonation, oxidation etc. go hand in hand and hasten the weathering process.

WEATHERING

2

PHYSICAL WEATHERING PROCESSES



1

Physical or mechanical weathering processes depend on some applied forces. The applied forces could be;

- i. **Gravitational forces** such as overburden pressure, load and shearing stress
- ii. **Expansion forces** due to temperature changes, crystal growth or animal activity
- iii. **Water pressures** controlled by wetting and drying cycles.

2

Many of these forces are applied both at the surface and within different earth materials leading to rock fracture.

WEATHERING

2

PHYSICAL WEATHERING PROCESSES



3

Most of the physical weathering processes are caused by thermal expansion and pressure release.

4

These processes are small and slow but can cause great damage to the rocks because of continued fatigue the rocks suffer due to repetition of contraction and expansion.

WEATHERING

2

PHYSICAL WEATHERING PROCESSES



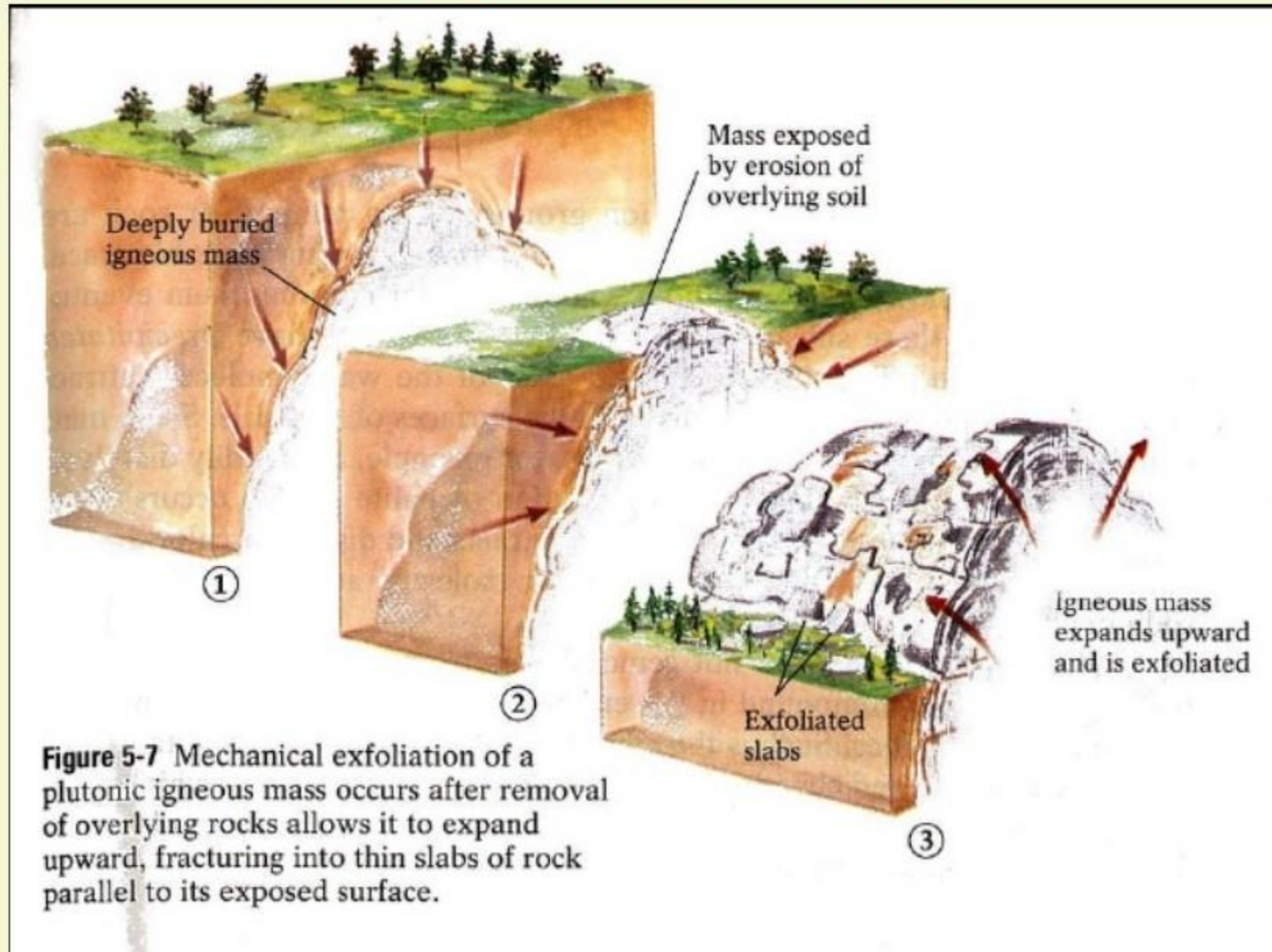
*Batholith of Mount Rushmore Hills,
USA exposed by unloading and
exfoliation*

UNLOADING AND EXPANSION

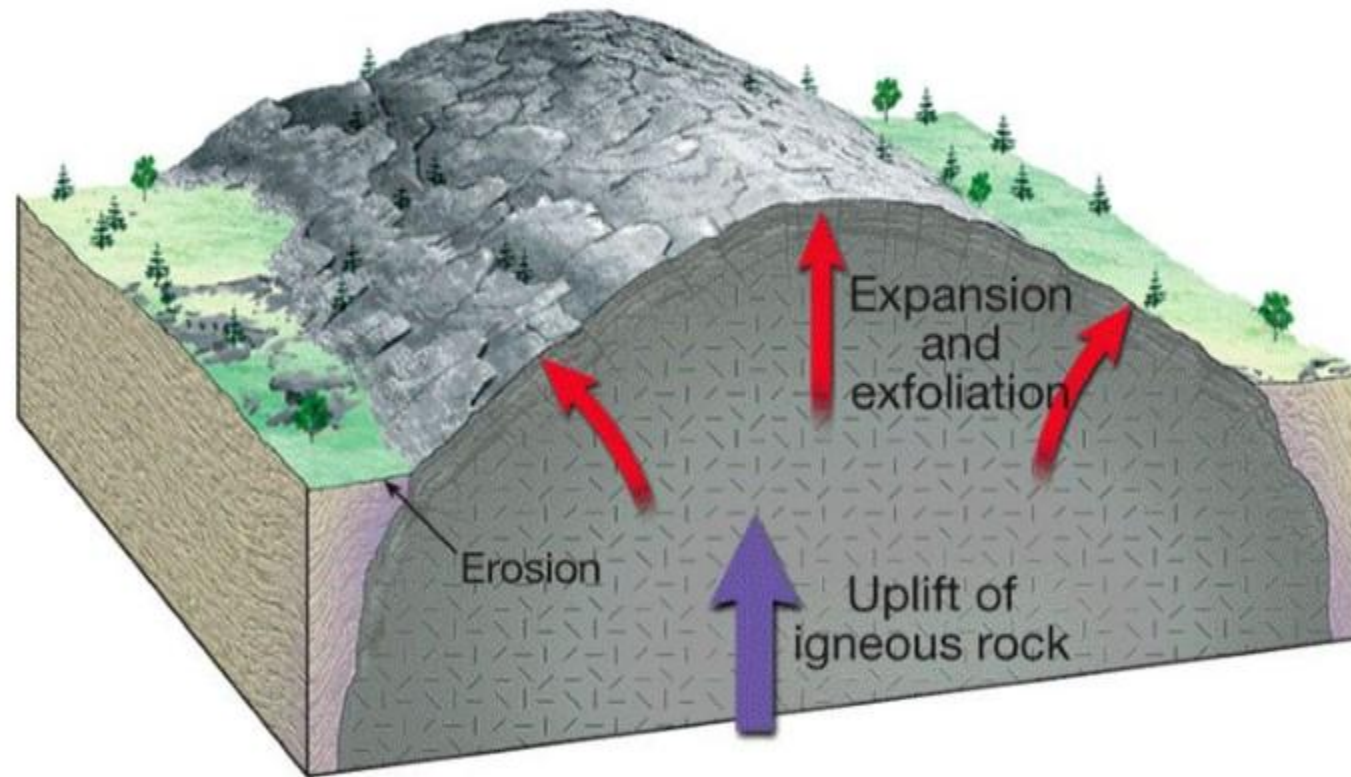
- 1 Removal of overlying rock load because of continued erosion causes vertical pressure release with the result that the upper layers of the rock expand producing disintegration of rock masses.
- 2 Fractures will develop roughly parallel to the ground surface. In areas of curved ground surface, arched fracture tend to produce massive sheets or exfoliation slabs of rock.

PHYSICAL WEATHERING PROCESSES

UNLOADING AND EXPANSION



Unloading and Exfoliation of Igneous Rocks



WEATHERING

2

PHYSICAL WEATHERING PROCESSES

UNLOADING AND EXPANSION



Large exfoliation domes created because of unloading and expansion

3

Exfoliation sheets resulting from expansion due to unloading and pressure release may measure hundreds or even thousands of metres in horizontal extent.

4

Large, smooth rounded domes called exfoliation domes result due to this process.

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